THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 01

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 299792458 m/s	a defined physical constant
2	**(b)** distance and average speed	the details that decide the time
3	**(a)** solid particles in a gas	solid dispersed in air
4	**(a)** suspension	particles settle on standing
5	**(a)** $MgCl_2$	metal + non-metal; electron transfer
6	**(c)** sugar is covalent and does not form ions	no free charge carriers
7	**(c)** 2, 8, 1	fill shells in order with maximum 2, 8, 8, 2
8	**(c)** 90 m	$v^2 = u^2 + 2as$
9	**(b)** a straight inclined line on a velocity–time graph	constant slope of v–t
10	**(b)** 12 N	F = ma
11	**(c)** 600 N	W = mg
12	**(a)** four times	KE ∝ v^2
13	**(c)** frequency	high frequency, high pitch
14	**(d)** 120 days	no nucleus, so no repair
15	**(c)** cancer cells	lack of control
16	**(a)** phloem cells	totipotency
17	**(b)** Kingdom, Phylum, Class, Order, Family, Genus, Species	broad to specific
18	**(d)** gametes from two parents combine	mixing of chromosomes
19	**(c)** meiosis	halves the chromosome number
20	**(a)** high pressure to low pressure	wind

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	It helps check whether a calculated answer is reasonable, build intuition and detect errors (2).	2
22	Neutral entity of more than one atom capable of independent existence (1); H_2 / O_2 and H_2O / HCl (1).	1+1
23	$v = 0 + 3 \times 4 = 12\ m\ s^{-1}$ (2).	2
24	Forces are equal and opposite but the gun is much heavier, so its acceleration is much less (a = F/m) (2).	2
25	Mitosis gives two identical cells for growth; meiosis gives four gametes with half the chromosomes (1+1).	1+1
26	Tendon connects muscle to bone; ligament connects bone to bone (1+1).	1+1
27	Transfer of pollen from anther to stigma (1); wind, water, insects, birds (1).	1+1
28	Evaporation, transpiration, condensation, precipitation, infiltration (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Milk particles are colloidal and scatter light (2); salt solution particles are too small to scatter (1).	2+1
30	Isotopes: same Z, different A; e.g. ^{12}C and ^{14}C (1.5). Isobars: same A, different Z; e.g. ^{40}Ar, ^{40}K, ^{40}Ca (1.5).	1½+1½
31	Work doubles (1.5); power is the same (W/t = 2W/2t) (1.5).	1½+1½
32	Medical imaging; breaking kidney stones; cleaning delicate parts; detecting flaws in metal blocks (1+1+1).	1+1+1
33	Bone: hard matrix with calcium compounds, strength; cartilage: soft, jelly-like matrix, flexible cushion (1.5+1.5).	1½+1½
34	Birds: feathers, wings, four-chambered heart; mammals: hair, mammary glands, live young (1.5+1.5).	1½+1½

35	When atoms combine the energy of the system becomes lower than that of the separate atoms (2), so the arrangement is more stable (1).	2+1
	OR: H_2SO_4: $2 + 32 + 64 = 98$ u (1.5); $CaCl_2$: $40 + 71 = 111$ u (1.5).	1½+1½
36	Oceans hold about 71% of the carbon and dissolve CO_2 (1.5); excess CO_2 acidifies oceans and harms marine life (1.5).	1½+1½
	OR: Artificial fixation of nitrogen to make ammonia fertilisers (1.5); boosted food production but is energy intensive and causes pollution (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) 12 : 32 = 3 : 8 (1). (ii) $\frac{12}{44} \times 22 = 6$ g (2). (iii) $\frac{8}{3} \times 6 = 16$ g (2).	1+2+2
	OR: (i) Zinc chloride and hydrogen gas (1). (ii) Hydrogen escaped (1). (iii) Use a closed flask with a balloon or stopper (2). (iv) Equal masses before and after (1).	1+1+2+1
38	(i) 2.5 m s^{-2} (1). (ii) 2000 N (1). (iii) 125 m (2). (iv) 500 m (1).	1+1+2+1
	OR: (i) Rate of change of momentum is proportional to net force (1). (ii) Product of mass and velocity (1). (iii) Δp = 10, F = 10/0.5 = 20 N (3).	1+1+3
39	(i) Orderly sequence of growth and division of a cell (1). (ii) Number of divisions, daughter cells, chromosome number, purpose (3). (iii) Growth, repair, reproduction (1).	1+3+1
	OR: (i) Growing cells outside the body (1). (ii) To prevent contamination by microbes (1). (iii) Production of biochemicals or study of cells (1). (iv) The ability of a plant cell to grow into a whole plant (2).	1+1+1+2
40	(i) Site, number of eggs, examples (2). (ii) The embryo is protected inside (1). (iii) Fertilisation in a dish, then the embryo placed in the uterus (2).	2+1+2
	OR: (i) Balanced diet rich in proteins, vitamins and minerals (1). (ii) Complete nutrition and protection from infection (1). (iii) Anxiety or low mood after delivery that can be treated (1). (iv) To prevent infections in mother and baby (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER